If it is determined that there are k factors that sufficiently explain returns, the risk exposures are determined from the first k risk exposures for each stock since our eigenvalues are sorted from highest predictive power to lowest.

$$\beta = \begin{bmatrix} \beta_{11} & \beta_{21} & \cdots & \beta_{m1} \\ \beta_{12} & \beta_{22} & \cdots & \beta_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ \beta_{1k} & \beta_{2k} & \cdots & \beta_{mk} \end{bmatrix}$$

The estimated covariance matrix is then:

$$C = \underset{nxk}{\beta'} \underset{kXn}{\beta} + \underset{nXn}{\Lambda} \tag{6.44}$$

In this case the idiosyncratic matrix Λ is the diagonal matrix consisting of the difference between the sample covariance matrix and $\beta'\beta$. That is,

$$\Lambda = diag(\overline{\mathrm{C}} - \beta'\beta) \tag{6.45}$$

It is important to note that in the above expression $\overline{\mathrm{C}} - \beta'\beta$ the off-diagonal terms will often be non-zero. This difference is considered to be the spurious relationship caused by the data limitation and degrees of freedom issue stated above. Selection of an appropriate number of factors determined via eigenvalue decomposition will help eliminate these false relationships.